

Longitudinal vibrations of a cylindrical rod based on the Rayleigh-Bishop theory

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Abstract A one dimensional model describing the longitudinal vibrations of a cylindrical rod composed of multiple sections with different diameters and made of different materials is derived. The hypothesis that lateral displacements and axial shear stresses in the bar are zero is rejected. The lateral displacements are assumed to be proportional to the axial strain and the Poisson ratio, resulting in the so-called Rayleigh-Bishop equation as opposed to the classical one dimensional wave equation. The Rayleigh-Bishop equation is a fourth order partial differential equation, not resolved with respect to the highest order time derivative. A system of partial differential equations, as well as the corresponding boundary conditions and continuity conditions at the junctions between the sections, are derived using Hamilton's variational principle. An analytical solution is obtained in terms of a Green function. A numerical example is provided for a rod consisting of two cylindrical sections, in which the first five eigenvalues and corresponding eigenfunctions are presented.

Keywords Longitudinal vibrations · Rayleigh-Bishop · Green function

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1 Introduction

Solid isotropic rods are often used for generating, transmitting and amplifying mechanical vibrations in, for example, acoustic transducers. Theoretical investigations of longitudinal

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vibrations in low frequency underwater acoustic transducers are generally based on the analysis of second order wave equations [2, 10, 11]. This approach is valid for the description of low frequency longitudinal wave propagation in long and thin solid rods. Rayleigh [8] showed that estimations made by the classical wave equation analysis could be improved on by taking the inertial effects due to the transverse deformations of the rod into account. Bishop [1] proposed that, in addition to the inertial effects, the effects of shear stress in the transverse cross section should also be taken into consideration. Inclusion of these effects when deriving the equation of motion results in the so called Rayleigh-Bishop equation [7], a linear fourth order partial differential equation not resolved with respect to the highest order time derivative. It has been shown [5, 9] that the Rayleigh-Bishop model improves on estimations made by the classical wave equation. The Rayleigh-Bishop model makes it possible to analyse longitudinal wave propagation in solid rods that are relatively thick (the maximum radius is comparable with the length of the rod) due to the inclusion of transverse effects in the model. Although the applicable range of vibration (or excitation) frequencies is still limited, analysis of higher frequency vibrations has also been justified [9].

The approach used in this work for the analysis of longitudinal vibrations of a rod consisting of multiple sections is based on the Rayleigh-Bishop model. Each section of the rod can be constructed of different materials and have different cross sectional areas. The ends of the rod, as well as the junction points where adjacent sections are joined, are fixed by means of lumped masses and springs. The equations of motion, and boundary and junction conditions are derived using Hamilton's variational principle. The differential equations are solved using the method of separation of variables, making it possible to construct the Green function. A numerical example of a solution for a rod consisting of two cylindrical sections, fixed at one end point and free at the junction point and other end point, is presented.

This publication originated from practical work done by one of the co-authors on the design of underwater acoustic transducers for South African corvettes. The authors are now busy with the creation of a general theory of mechanical wave guides. This paper is one of their first publications in this field. An analogous approach has been proposed in an earlier paper [4], but with the longitudinal vibrations based on the so-called Rayleigh-Love model.

Free, harmonic vibrations of the rod have been considered in this work.

2 Formulation of the problem

Consider a solid rod composed of n cylindrical sections made of different, isotopic materials, which experiences longitudinal vibrations in the direction of the x axis. The rod can be fixed at the end points and/or at any one of the junctions between the sections by means of lumped masses and springs. In Fig. 1, these are represented by lumped masses with mass M_j and springs with stiffness K_j ($j = 0, 1, 2, \dots, n$). The purpose of these lumped masses and springs is to simulate semi-rigid supports used to fix the ends (or junctions) of the rod.

The longitudinal displacement in each section of the rod is designated as $u_j = u_j(x, t)$, ($j = 1, 2, \dots, n$). As in the model proposed by Rayleigh, the lateral displacement in each section of the rod is designated as $w_j = -r_j \eta_j u'_j(x, t)$, ($j = 1, 2, \dots, n$), where the prime denotes the derivative with respect to x , r_j represents the radial coordinate and η_j is the Poisson ratio of section j of the rod. That is, the displacement of a particle distant r from the longitudinal axis is assumed to be proportional to the longitudinal strain. Applying this constraint to the lateral displacement means that the normal radial stress, σ_{rr} , is equal to zero throughout the entire thickness of the bar.

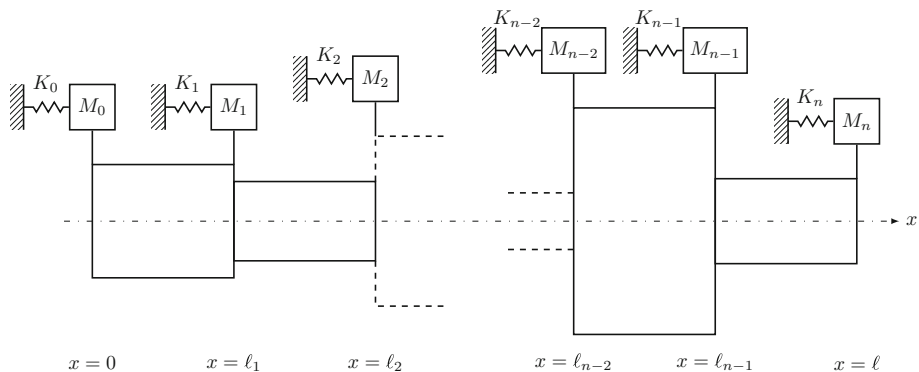


Fig. 1 Schematic representation of an n -stepped bar

In this way, one of the boundary conditions for a free outer surface in two or three dimensional models, $\sigma_{rr}|_{r=R} = 0$, is implicitly taken into account.

2.1 Equations of motion and boundary conditions

Equations of motion and boundary conditions are derived using Hamilton's variational principle. The variational approach to derivation of equations and boundary conditions is demonstrated by Rao [6]. The Lagrangian for a Rayleigh-Bishop rod consisting of n sections and fixed by means of lumped masses and springs can be written as

$$\mathcal{L} = \mathcal{L}(u_j, u'_j, \dot{u}_j, \dot{u}'_j, u''_j, u_j(\ell_j, t), \dot{u}_j(\ell_j, t)) = T - P, \quad (1)$$

where the upper dot denotes a derivative with respect to t . In Eq. (1),

$$T = \sum_{j=1}^n \frac{1}{2} \left[\rho_j S_j \dot{u}_j^2 + \rho_j \eta_j^2 I_j (\dot{u}'_j)^2 \right] + \sum_{j=0}^n \frac{1}{2} M_j \dot{u}_j^2(\ell_j, t) \quad (2)$$

is the kinetic energy yielded (or supplied) by the displacement of the disturbances (vibrations) in the bar,

$$P = \sum_{j=1}^n \frac{1}{2} \left[E_j S_j (u'_j)^2 + \mu_j \eta_j^2 I_j (u''_j)^2 \right] + \sum_{j=0}^n \frac{1}{2} K_j u_j^2(\ell_j, t) \quad (3)$$

is the sum of the potential (or strain) energy stored in each section of the rod as a result of elastic straining, $u_0 := u_1$, $\ell_0 := 0$, $\ell_n := \ell$, $S_j = \iint_S dS$ is the cross sectional area of section j of the rod, $I_j = \iint_S r_j^2 dS$ is the polar moment of inertia, ρ_j is the mass density, E_j is the modulus of elasticity, η_j is the Poisson ratio and $\mu_j = \frac{E_j}{2(1+\eta_j)}$ is the shear modulus.

Applying Hamilton's variational principle to the Lagrangian (1) requires that the so-called action integral be set equal to zero:

$$\int_{t_1}^{t_2} \delta \mathcal{L} dt = 0, \quad (4)$$

where t_1 and t_2 are arbitrary. This results in the following system of n homogeneous partial differential equations

$$\rho_j S_j \frac{\partial^2 u_j}{\partial t^2} = E_j S_j \frac{\partial^2 u_j}{\partial x^2} + \rho_j \eta_j^2 I_j \frac{\partial^4 u_j}{\partial x^2 \partial t^2} - \mu_j \eta_j^2 I_j \frac{\partial^4 u_j}{\partial x^4}, \quad (j = 1, 2, \dots, n), \quad (5)$$

subject to the following natural boundary and junction conditions

$$\begin{aligned} x = 0 : & \quad \begin{cases} \Gamma_1 u_1|_0 + K_0 u_1(0, t) + M_0 \ddot{u}_1(0, t) = 0, \\ u_1''(0, t) = 0, \end{cases} \\ x = \ell_j : & \quad \begin{cases} \Gamma_j u_j|_{\ell_j} = \Gamma_{j+1} u_{j+1}|_{\ell_j} + K_j u_j(\ell_j, t) + M_j \ddot{u}_j(\ell_j, t), \\ \mu_j \eta_j^2 I_j u_j''(\ell_j, t) = \mu_{j+1} \eta_{j+1}^2 I_{j+1} u_{j+1}''(\ell_j, t), \\ u_{j+1}(\ell_j, t) = u_j(\ell_j, t), \\ u_{j+1}'(\ell_j, t) = u_j'(\ell_j, t), \\ (j = 1, 2, \dots, n-1), \end{cases} \\ x = \ell : & \quad \begin{cases} \Gamma_n u_n|_{\ell} - K_n u_n(\ell, t) - M_n \ddot{u}_n(\ell, t) = 0, \\ u_n''(\ell, t) = 0, \end{cases} \end{aligned} \quad (6)$$

The differential operator Γ_j is defined as

$$\Gamma_j : C^{(2)}[0, \tau] \times C^{(3)}[\ell_{j-1}, \ell_j] \ni u_j(x, t) \rightarrow C^{(0)}[0, \tau] \times C^{(0)}[\ell_{j-1}, \ell_j] :$$

$$\Gamma_j u_j = \left[-E_j S_j \frac{\partial}{\partial x} - \rho_j \eta_j^2 I_j \frac{\partial^3}{\partial x \partial t^2} + \mu_j \eta_j^2 I_j \frac{\partial^3}{\partial x^3} \right] u_j(x, t), \quad (7)$$

and the restriction of this operator on $x = \ell_j$

$$\begin{aligned} \Gamma_j u_j|_{\ell_j} = & -E_j S_j \left(\frac{\partial}{\partial x} u_j(x, t) \right)_{\ell_j} - \rho_j \eta_j^2 I_j \left(\frac{\partial^3}{\partial x \partial t^2} u_j(x, t) \right)_{\ell_j} \\ & + \mu_j \eta_j^2 I_j \left(\frac{\partial^3}{\partial x^3} u_j(x, t) \right)_{\ell_j}. \end{aligned} \quad (8)$$

In the boundary conditions (6), $\Gamma_j u_j|_{\ell_j}$ represents the traction forces at the cross section denoted by $u_j(\ell_j, t)$, and $K_j u_j(\ell_j, t)$ and $M_j \ddot{u}_j(\ell_j, t)$ represent the stiffness force of the spring and the inertial force of the mass located at $x = \ell_j$ respectively (where $j = 0, 1, 2, \dots, n$ and $u_0 := u_1$). The second boundary conditions at $x = 0$ and $x = \ell_n$ require that the axial shear stress be equal to zero at the end points. The third and fourth conditions at the junction points $x = \ell_j$ ($j = 1, 2, \dots, n-1$) represent (respectively) the requirement for the solution to be continuous, and a special requirement for the necessary smoothness of the solution at $x = \ell_j$.

The following combinations of boundary conditions are possible, all of which satisfy Hamilton's variational principle:

- As described by (6). In this case, the end points and junction points are fixed in the longitudinal direction by means of semi-rigid supports simulated by means of lumped masses M_j and springs K_j . The end points are free to move in the lateral direction in a manner similar to roller-type conditions, and the junction points are fixed in the lateral

- direction in a manner similar to encastre-type conditions (the cross sections of adjacent sections cannot move independently of each other in the lateral direction).
- The second and third conditions at the junction points $x = \ell_j$ ($j = 1, 2, \dots, n - 1$) in (6) are replaced by the conditions $u_j''(\ell_j, t) = u_{j+1}''(\ell_j, t) = 0$. In this case, the junction points are free to move (independently of each other) in a manner similar to roller-type conditions.
 - The boundary conditions corresponding to free (in the longitudinal direction) end points and junction points are obtained by setting $K_j = M_j = 0$, ($j = 0, 1, 2, \dots, n$) in (6).
 - The essential conditions for end points and junction points that are fixed in both the longitudinal and lateral directions (encastre-type conditions), are obtained by replacing the boundary conditions in (6) with $u_1(0, t) = u_1'(0, t) = 0$, $u_{j+1}(\ell_j, t) = u_j(\ell_j, t) = 0$, $u_{j+1}'(\ell_j, t) = u_j'(\ell_j, t) = 0$, ($j = 1, 2, \dots, n - 1$), and $u_n(\ell_n, t) = u_n'(\ell_n, t) = 0$.
 - The essential conditions corresponding to end points and junction points that are fixed in the longitudinal direction, but free to move in the lateral direction (roller type conditions) are obtained by replacing the boundary conditions in (6) by $u_1(0, t) = u_1'(0, t) = 0$, $u_{j+1}(\ell_j, t) = u_j(\ell_j, t) = 0$, $u_{j+1}''(\ell_j, t) = u_j''(\ell_j, t) = 0$, ($j = 1, 2, \dots, n - 1$), and $u_n(\ell_n, t) = u_n''(\ell_n, t) = 0$.

It should be noted that the special condition relating to the smoothness of the global solution at the junction points, $u_{j+1}'(\ell_j, t) = u_j'(\ell_j, t)$, is rejected with the roller type junction conditions.

The partial differential equations (5), subject to the boundary and junction conditions (6) and the initial conditions

$$u_j(x, 0) = \varphi_j(x), \quad \dot{u}_j(x, 0) = \psi_j(x), \quad (j = 1, 2, \dots, n). \quad (9)$$

are considered in the sections that follow.

3 The Sturm-Liouville problem

Suppose that the solution of the homogeneous partial differential equations (5) can be described by

$$u_j(x, t) = y_j(x) \Phi(t) \quad (10)$$

Substituting (10) into the PDE's (5) gives n ordinary differential equations

$$E_j S_j y_j''(x) - \mu_j \eta_j^2 I_j y_j^{(\text{iv})}(x) + \lambda \rho_j S_j y_j(x) - \lambda \rho_j \eta_j^2 I_j y_j''(x) = 0. \quad (11)$$

Thus $y_j(x) \in C^{(\text{iv})}[\ell_{j-1}, \ell_j]$.

Before formulating the boundary conditions, it is convenient to introduce the operator Γ_j' :

$$\Gamma_j'(\lambda, \alpha) : C^{(3)}[\ell_{j-1}, \ell_j] \ni y_j(x) \rightarrow \mathbb{R}, \quad \alpha \in [\ell_{j-1}, \ell_j] :$$

$$\Gamma_j'(\lambda, \alpha) y_j = [-E_j S_j + \lambda \rho_j \eta_j^2 I_j] y_j'(\alpha) + \mu_j \eta_j^2 I_j y_j'''(\alpha), \quad (12)$$

so that the definition of the restriction of operator Γ_j on $x = \ell_j$ given by Eq. (8) can be written as

$$\Gamma_j u_j \Big|_{\ell_j} = \Phi(t) \Gamma_j'(\lambda, \ell_j) y_j.$$

Thus, substituting (10) into the boundary conditions (6) gives

$$\begin{aligned}
 x = 0 : \quad & \begin{cases} \Gamma'_1(\lambda, 0)y_1 + K_0 y_1(0) - \lambda M_0 y_1(0) = 0 \\ y'_1(0) = 0, \end{cases} \\
 x = \ell_j : \quad & \begin{cases} \Gamma'_j(\lambda, \ell_j)y_j = \Gamma'_{j+1}(\lambda, \ell_j)y_{j+1} + K_j y_j(\ell_j) - \lambda M_j y_j(\ell_j) \\ \mu_j \eta_j^2 I_j y''_j(\ell_j) = \mu_{j+1} \eta_{j+1}^2 I_{j+1} y''_{j+1}(\ell_j), \\ y_{j+1}(\ell_j) = y_j(\ell_j), \\ y'_{j+1}(\ell_j) = y'_j(\ell_j), \\ (j = 1, 2, \dots, n-1), \end{cases} \\
 x = \ell : \quad & \begin{cases} \Gamma'_n(\lambda, \ell)y_n - K_n y_n(\ell) + \lambda M_n y_n(\ell) = 0, \\ y'_n(\ell) = 0, \end{cases}
 \end{aligned} \tag{13}$$

3.1 Orthogonality of the eigenfunctions

Let

$$y(x) = \sum_{j=1}^n y_j(x) \Delta H_j(x) \quad \text{and} \quad z(x) = \sum_{j=1}^n z_j(x) \Delta H_j(x), \tag{14}$$

where $\Delta H_j(x) = H(x - \ell_{j-1}) - H(x - \ell_j)$ and $H(x)$ is the Heaviside or unit step function, be two distinct, arbitrary real valued eigenfunctions corresponding to the positive, distinct eigenvalues λ_y and λ_z respectively, and each satisfying the n ordinary differential equations

$$(A_j + \lambda_y B)y_j = 0, \tag{15}$$

and

$$(A_j + \lambda_z B)z_j = 0, \tag{16}$$

respectively. The differential operators A_j and B_j are defined as

$$\begin{aligned}
 A_j y_j &= -E_j S_j \frac{d^2}{dx^2} y_j(x) + \mu_j \eta_j^2 I_j \frac{d^4}{dx^4} y_j(x), \\
 B_j y_j &= \rho_j \eta_j^2 I_j \frac{d^2}{dx^2} y_j(x) - \rho_j S_j y_j(x).
 \end{aligned}$$

In what follows, two different orthogonality conditions of the eigenfunctions (14) will be presented.

Theorem 1 (First condition of orthogonality) *Let $y(x) = \sum_{j=1}^n y_j(x) \Delta H_j(x) \in C^{(1)}[0, \ell]$ and $z(x) = \sum_{j=1}^n z_j(x) \Delta H_j(x) \in C^{(1)}[0, \ell]$, where $y_j(x) \in C^{(4)}[\ell_{j-1}, \ell_j]$ and $z_j(x) \in C^{(4)}[\ell_{j-1}, \ell_j]$, be two eigenfunctions of the Sturm-Liouville problem (11)–(13), corresponding to the eigenvalues λ_y and λ_z respectively. If λ_y and λ_z are distinct, then the eigenfunctions $y(x)$ and $z(x)$ are orthogonal in the following sense:*

$$\sum_{j=1}^n \int_{\ell_{j-1}}^{\ell_j} [\rho_j S_j y_j(x) z_j(x) + \rho_j \eta_j^2 I_j y'_j(x) z'_j(x)] dx + \sum_{j=0}^n M_j y_j(\ell_j) z_j(\ell_j) = 0.$$

Proof Take the sum (from $j = 1$ to n) of all n scalar products of equation (15) and z_j , and subtract from this the sum (from $j = 1$ to n) of the scalar products of equation (16) and y_j :

$$\sum_{j=1}^n \langle A_j y_j, z_j \rangle - \sum_{j=1}^n \langle A_j z_j, y_j \rangle + \lambda_y \sum_{j=1}^n \langle B_j y_j, z_j \rangle - \lambda_z \sum_{j=1}^n \langle B_j z_j, y_j \rangle = 0. \quad (17)$$

Using integration by parts, it is possible to write

$$\begin{aligned} \langle A_j y_j, z_j \rangle &= -E_j S_j z_j(x) y_j'(x) \Big|_{\ell_{j-1}}^{\ell_j} + \mu_j \eta_j^2 I_j z_j(x) y_j'''(x) \Big|_{\ell_{j-1}}^{\ell_j} \\ &\quad - \mu_j \eta_j^2 I_j z_j'(x) y_j''(x) \Big|_{\ell_{j-1}}^{\ell_j} \\ &\quad + \int_{\ell_{j-1}}^{\ell_j} E_j S_j y_j'(x) z_j'(x) dx + \int_{\ell_{j-1}}^{\ell_j} \mu_j \eta_j^2 I_j y_j''(x) z_j''(x) dx, \\ \langle B_j y_j, z_j \rangle &= -\rho_j \eta_j^2 I_j z_j(x) y_j'(x) \Big|_{\ell_{j-1}}^{\ell_j} - \int_{\ell_{j-1}}^{\ell_j} \rho_j \eta_j^2 I_j y_j'(x) z_j'(x) dx \\ &\quad - \int_{\ell_{j-1}}^{\ell_j} \rho_j S_j y_j(x) z_j(x) dx. \end{aligned}$$

Similar formulae can be found for $\langle A_j z_j, y_j \rangle$ and $\langle B_j z_j, y_j \rangle$. After making some transformations and applying the boundary conditions, Eq. (17) can be written as

$$\begin{aligned} (\lambda_z - \lambda_y) \left\{ \sum_{j=1}^n \int_{\ell_{j-1}}^{\ell_j} [\rho_j S_j y_j(x) z_j(x) + \rho_j \eta_j^2 I_j y_j'(x) z_j'(x)] dx \right. \\ \left. + \sum_{j=0}^n M_j y_j(\ell_j) z_j(\ell_j) \right\} = 0. \end{aligned}$$

Since the eigenvalues λ_z and λ_y are distinct, the first orthogonality condition is obtained as

$$\sum_{j=1}^n \int_{\ell_{j-1}}^{\ell_j} [\rho_j S_j y_j(x) z_j(x) + \rho_j \eta_j^2 I_j y_j'(x) z_j'(x)] dx + \sum_{j=0}^n M_j y_j(\ell_j) z_j(\ell_j) = 0. \quad (18)$$

□

Note that the standard notation for scalar product in the real space $L_2[p, q]$ is implied:

$$\langle u, v \rangle = \langle u(x), v(x) \rangle = \int_p^q u(x) v(x) dx.$$

The square norm corresponding to the first orthogonality condition (18) can be written as

$$\|y\|_1^2 = \sum_{j=1}^n \int_{\ell_{j-1}}^{\ell_j} [\rho_j S_j y_j^2(x) + \rho_j \eta_j^2 I_j (y_j'(x))^2] dx + \sum_{j=0}^n M_j y_j^2(\ell_j). \quad (19)$$

Theorem 2 (Second condition of orthogonality) *Let $y(x) = \sum_{j=1}^n y_j(x) \Delta H_j(x) \in C^{(1)}[0, \ell]$ and $z(x) = \sum_{j=1}^n z_j(x) \Delta H_j(x) \in C^{(1)}[0, \ell]$, where $y_j(x) \in C^{(4)}[\ell_{j-1}, \ell_j]$*

and $z_j(x) \in C^{(4)}[\ell_{j-1}, \ell_j]$, be two eigenfunctions of the Sturm-Liouville problem (11)–(13), corresponding to the eigenvalues λ_y and λ_z respectively. If λ_y and λ_z are distinct, then the eigenfunctions $y(x)$ and $z(x)$ are orthogonal in the following sense:

$$\sum_{j=1}^n \int_{\ell_{j-1}}^{\ell_j} [E_j S_j y_j'(x) z_j'(x) + \mu_j \eta_j^2 I_j y_j''(x) z_j''(x)] dx + \sum_{j=0}^n K_j y_j(\ell_j) z_j(\ell_j) = 0. \quad (20)$$

The proof of Theorem 2 is obtained in a manner similar to that for Theorem 1, by taking the sum (from $j = 1$ to n) of all n scalar products of Eq. (15) and $\lambda_z z_j$, and subtracting from this the sum (from $j = 1$ to n) of the scalar products of Eq. (16) and $\lambda_y y_j$:

$$\lambda_z \sum_{j=1}^n \langle A_j y_j, z_j \rangle - \lambda_y \sum_{j=1}^n \langle A_j z_j, y_j \rangle + \lambda_z \lambda_y \sum_{j=1}^n \langle B_j y_j, z_j \rangle - \lambda_y \lambda_z \sum_{j=1}^n \langle B_j z_j, y_j \rangle = 0,$$

and making use of the boundary and junction conditions (13).

The square (semi) norm corresponding to the second orthogonality condition (20) can be written as

$$\|y\|_2^2 = \sum_{j=1}^n \int_{\ell_{j-1}}^{\ell_j} [E_j S_j (y_j'(x))^2 + \mu_j \eta_j^2 I_j (y_j''(x))^2] dx + \sum_{j=0}^n K_j y_j^2(\ell_j). \quad (21)$$

3.2 Nature of the eigenvalues

Theorem 3 (Positivity of the eigenvalues) *If E_j , ρ_j , η_j , μ_j , S_j and I_j are all positive, and $y(x) = \sum_{j=1}^n y_j(x) \Delta H_j(x) \in C^{(1)}[0, \ell]$ is an eigenfunction of the Sturm-Liouville problem (11), (13), where $y_j(x) \in C^{(4)}[\ell_{j-1}, \ell_j]$, then the eigenvalue, λ , corresponding to $y(x)$ is positive. Moreover,*

$$\lambda = \frac{\sum_{j=1}^n \int_{\ell_{j-1}}^{\ell_j} [E_j S_j (y_j'(x))^2 + \mu_j \eta_j^2 I_j (y_j''(x))^2] dx + \sum_{j=0}^n K_j y_j^2(\ell_j)}{\sum_{j=1}^n \int_{\ell_{j-1}}^{\ell_j} [\rho_j S_j y_j^2(x) + \rho_j \eta_j^2 I_j (y_j'(x))^2] dx + \sum_{j=0}^n M_j y_j^2(\ell_j)} = \frac{\|y\|_2^2}{\|y\|_1^2}.$$

The proof of Theorem 3 is obtained in a manner similar to that for Theorem 1, by taking the sum (from $j = 1$ to n) of all n scalar products of equation (15) and y_j :

$$\sum_{j=1}^n \langle A_j y_j, y_j \rangle + \lambda \sum_{j=1}^n \langle B_j y_j, y_j \rangle = 0,$$

and making use of the boundary and junction conditions (13).

Since the eigenvalues are positive and distinct, it is convenient to write the eigenvalues as

$$\lambda_k = \omega_k^2, \quad (k = 1, 2, \dots, \infty),$$

and the eigenfunction corresponding to ω_k as

$$y(\omega_k, x) = \sum_{j=1}^n y_j(\omega_k, x) \Delta H_j(x),$$

where $y_j(\omega_k, x) \in C^{(4)}[\ell_{j-1}, \ell_j]$ satisfies the differential equation $(A_j + \omega_k^2 B_j)y_j = 0$.

4 Numerical example: eigenvalues and eigenfunctions

Consider free harmonic vibrations of a rod consisting of two sections with constant cross-sectional areas. In this case, if the radius of each section is R_1 and R_2 , then the cross-sectional areas and polar moments of inertia of the two sections are determined from the formulae

$$S_1 = \pi R_1^2, \quad I_1 = \frac{1}{2} \pi R_1^4, \quad S_2 = \pi R_2^2, \quad I_2 = \frac{1}{2} \pi R_2^4.$$

The general solution of the corresponding Sturm-Liouville problem is given by

$$y_1(\omega_k, x) = C_{11} \cos[k_{11}(\omega_k) x] + C_{12} \sin[k_{11}(\omega_k) x] + C_{13} \cosh[k_{12}(\omega_k) x] + C_{14} \sinh[k_{12}(\omega_k) x], \quad (22)$$

for the first section, and

$$y_2(\omega_k, x) = C_{21} \cos[k_{21}(\omega_k) x] + C_{22} \sin[k_{21}(\omega_k) x] + C_{23} \sinh[k_{22}(\omega_k) x] + C_{24} \cosh[k_{22}(\omega_k) x], \quad (23)$$

for the second section, where

$$k_{j1,2}(\omega_k) = \sqrt{\sqrt{\left(\frac{E_j S_j - \rho_j \omega_k^2 \eta_j^2 I_j}{2 \mu_j \eta_j^2 I_j}\right) + \frac{\rho_j \omega_k^2 S_j}{\mu_j \eta_j^2 I_j}} \mp \left(\frac{E_j S_j - \rho_j \omega_k^2 \eta_j^2 I_j}{2 \mu_j \eta_j^2 I_j}\right)}, \quad (24)$$

$(j = 1, 2).$

The rod is fixed at $x = 0$ so that both the longitudinal and lateral displacements are zero, and is free to move at the junction between the two sections at $x = \ell_1$ and at $x = \ell_2 = \ell$. The boundary and junction conditions corresponding to this problem can be given by

$$\begin{aligned} x = 0 : & \begin{cases} y_1(\omega_k, 0) = 0, \\ y_1'(\omega_k, 0) = 0, \end{cases} \\ x = \ell_1 : & \begin{cases} \Gamma_1'(\omega_k^2, \ell_1) y_1 = \Gamma_2'(\omega_k^2, \ell_1) y_2 \\ \mu_1 \eta_1^2 I_1 y_1''(\omega_k, \ell_1) = \mu_2 \eta_2^2 I_2 y_2''(\omega_k, \ell_1), \\ y_2(\omega_k, \ell_1) = y_1(\omega_k, \ell_1), \\ y_2'(\omega_k, \ell_1) = y_1'(\omega_k, \ell_1), \end{cases} \\ x = \ell : & \begin{cases} \Gamma_2'(\omega_k^2, \ell) y_2 = 0, \\ y_2''(\omega_k, \ell) = 0, \end{cases} \end{aligned} \quad (25)$$

Suppose further that both sections of the rod are composed of the same aluminium alloy and have different radii. The physical and geometrical parameters of the rod are as follows:

$$E_1 = E_2 = 70 \times 10^9 \text{ Pa}, \quad \rho_1 = \rho_2 = 2700 \text{ kg m}^{-3}, \quad \eta_1 = \eta_2 = 0.33, \\ R_1 = 0.11 \text{ m}, \quad R_2 = 0.138 \text{ m}, \quad \ell_1 = 0.20 \text{ m}, \quad \ell = 0.30 \text{ m}.$$

Substituting the general solutions (22) and (23) into the boundary conditions (25) results in a system of homogeneous linear equations for the coefficients C_{11} through C_{24} . The determinant of the resulting coefficient matrix is set equal to zero, which gives the dispersion equation $D(\omega_k) = 0$ that can be solved for the eigenvalues ω_k . The first five eigenvalues for

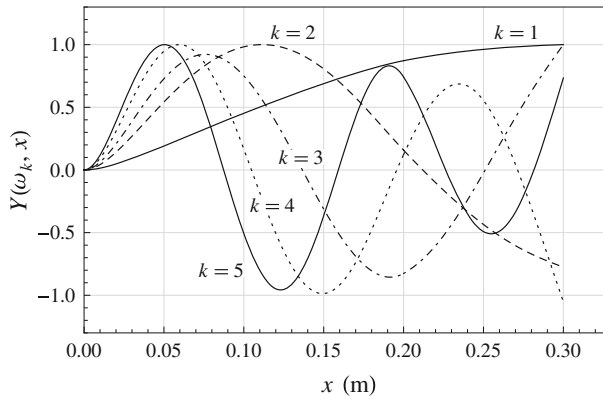


Fig. 2 A plot of the eigenfunctions corresponding to the first five eigenvalues

this case are given below (first four digits, in Hz)

$$\begin{aligned} \frac{\omega_1}{2\pi} &= 3896 \text{ Hz}, & \frac{\omega_2}{2\pi} &= 12710 \text{ Hz}, & \frac{\omega_3}{2\pi} &= 19040 \text{ Hz}, \\ \frac{\omega_4}{2\pi} &= 24400 \text{ Hz}, & \frac{\omega_5}{2\pi} &= 28810 \text{ Hz}. \end{aligned}$$

The normalised eigenfunctions

$$Y(\omega_k, x) = \frac{y(\omega_k, x)}{\max_{0 \leq x \leq \ell} |y(\omega_k, x)|},$$

corresponding to the first five eigenvalues are shown in Fig. 2. All numerical results and graphs were generated using Mathematica[®].

Fedotov et al. [5] showed that the eigenfunctions corresponding to the Sturm-Liouville problem (11)–(13) for a Rayleigh-Bishop rod with conical cross sections can be given in terms of generalised hypergeometric functions.

5 Solution of the BVP and the green function

Suppose that the solution of the partial differential equations (5), subject to the boundary and initial conditions (6) and (9) can be written in the form of a Fourier series expansion

$$u(x, t) = \sum_{k=1}^N [a_k \cos(\omega_k t) + b_k \sin(\omega_k t)] y(\omega_k, x), \quad (26)$$

where $y(\omega_k, x)$ is an eigenfunction corresponding to the eigenvalue ω_k , and a_k and b_k are constants that must be determined.

Substituting (26) into the initial conditions (9) gives

$$\varphi(x) = \sum_{k=1}^N a_k y(\omega_k, x), \quad \psi(x) = \sum_{k=1}^N b_k \omega_k y(\omega_k, x). \quad (27)$$

Using the first orthogonality condition (18) it is possible to determine the coefficients

$$a_k = \frac{1}{\|y(\omega_k, x)\|_1^2} \left\{ \sum_{j=1}^n \int_{\ell_{j-1}}^{\ell_j} [\rho_j S_j \varphi(\xi) y_j(\omega_k, \xi) + \rho_j \eta_j^2 I_j \varphi'(\xi) y_j'(\omega_k, \xi)] dx + \sum_{j=0}^n M_j \varphi(\ell_j) y_j(\omega_k, \ell_j) \right\}, \quad (28)$$

and

$$b_k = \frac{1}{\omega_k \|y(\omega_k, x)\|_1^2} \left\{ \sum_{j=1}^n \int_{\ell_{j-1}}^{\ell_j} [\rho_j S_j \psi(\xi) y_j(\omega_k, \xi) + \rho_j \eta_j^2 I_j \psi'(\xi) y_j'(\omega_k, \xi)] dx + \sum_{j=0}^n M_j \psi(\ell_j) y_j(\omega_k, \ell_j) \right\}, \quad (29)$$

where $y_0 := y_1$. Equations (28) and (29) are substituted into Eq. (26). After certain transformations, and making use of the following relations

$$\rho(x) = \sum_{j=1}^n \rho_j \Delta H_j(x), \quad \eta(x) = \sum_{j=1}^n \eta_j \Delta H_j(x), \quad S(x) = \sum_{j=1}^n S_j \Delta H_j(x), \\ I(x) = \sum_{j=1}^n I_j \Delta H_j(x), \quad y(\omega_k, x) = \sum_{j=1}^n y_j(\omega_k, x) \Delta H_j(x),$$

and

$$\alpha(x) = \rho(x) S(x), \quad \beta(x) = \rho(x) \eta^2(x) I(x),$$

the solution of the problem (5), (6), (9) corresponding to free harmonic vibrations of a Rayleigh-Bishop rod consisting of n cylindrical sections can be obtained as

$$u(x, t) = \sum_{j=0}^n M_j \varphi(\ell_j) \frac{\partial G}{\partial t}(x, \ell_j, t) + \sum_{j=0}^n M_j \psi(\ell_j) G(x, \ell_j, t) \\ + \int_0^\ell [\alpha(\xi) \varphi(\xi) \frac{\partial G}{\partial t}(x, \xi, t) + \beta(\xi) \varphi'(\xi) \frac{\partial^2 G}{\partial \xi \partial t}(x, \xi, t)] d\xi \\ + \int_0^\ell [\alpha(\xi) \psi(\xi) G(x, \xi, t) + \beta(\xi) \psi'(\xi) \frac{\partial G}{\partial \xi}(x, \xi, t)] d\xi, \quad (30)$$

where

$$G(x, \xi, t) = \sum_{k=1}^N \frac{1}{\omega_k \|y(\omega_k, x)\|_1^2} y(\omega_k, x) y(\omega_k, \xi) \sin(\omega_k t) \quad (31)$$

is the Green function. It is possible to show [3] that an alternative representation of the solution can be obtained by using the second orthogonality condition (20).

6 Conclusion

In this article, a one dimensional model describing free, harmonic, longitudinal vibrations of a rod consisting of n cylindrical sections was derived. Each section of the rod could be

constructed of different materials and have different cross sectional areas. The equations of motion and boundary and junction conditions were derived using Hamilton's variational principle. The lateral displacements of the rod were not neglected as in the case of the classical wave equation, but rather were assumed to be proportional to the longitudinal strain. As a result of this constraint, it was found that the displacements $u_j(x, t)$ ($j = 1, 2, \dots, n$) are required to satisfy a fourth order partial differential equation, the so-called Rayleigh-Bishop equation, in each section of the rod. The eigenfunctions were shown to satisfy two orthogonality conditions and the square norms corresponding to the orthogonality conditions were also given. A numerical example, in which eigenvalues and eigenfunctions were calculated for a rod composed of two aluminium sections with different radii, was presented. Finally, an analytical solution of the corresponding boundary value problem was derived in terms of the first orthogonality condition and Green functions.

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